



AtmoSound

An ML Framework for Venue-Adaptive Playlist Generation

Ridge Regression

Neural Network

91,271 Spotify Tracks

4,484 Venues

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The Problem with In-Store Music

THE PROBLEM

Most stores select playlists through intuition or B2B services (e.g., Soundtrack Your Brand, Mood Media) that deliver standardized genre channels
— >not aligned to a **specific venue's identity**.



Playlist for Restaurants



WHY IT MATTERS

- ▶ Background music affects **dwell time & spending**
E.g. Milliman: slower tempo → increased in-store time and spend
- ▶ Acoustic congruence drives **higher customer satisfaction**
- ▶ **Increase brand perception**

Prior Work & Research Gap in Music Recommendation

PHASE 1 — Collaborative Filtering

- Listening history
- Item co-occurrence

Challenge:

Inapplicable to venues — no individual listening history exists.

PHASE 2 — Content-Based & Hybrid

- Item Profiles Similarity

Van den Oord et al. (2013): CNNs on audio spectrograms → cold-start rec.

Wang & Wang (2014): hybrid audio+CF.

Challenge:

Optimize for individual preference — wrong objective for physical venues.

PHASE 3 — Context-Aware

- Dynamic Adaptation

Murciego et al. (2021): 128 papers

Kaminskas (2013): audio tags for POIs.

CAME [9]: contextual embeddings

Challenge:

Venue atmosphere = least studied.
Still requires user-item data

RESEARCH GAP — HOW ATMOSOUND DIFFERS

- Input: Google Maps venue metadata — not audio features or listening history. **No user data required.**
- Two-stage architecture: predict audio profile → retrieve songs. Retrieval updatable (live Spotify API) without retraining the model.

AtmoSound: What We Built

Google Maps URL



Venue Metadata



279-dim Features



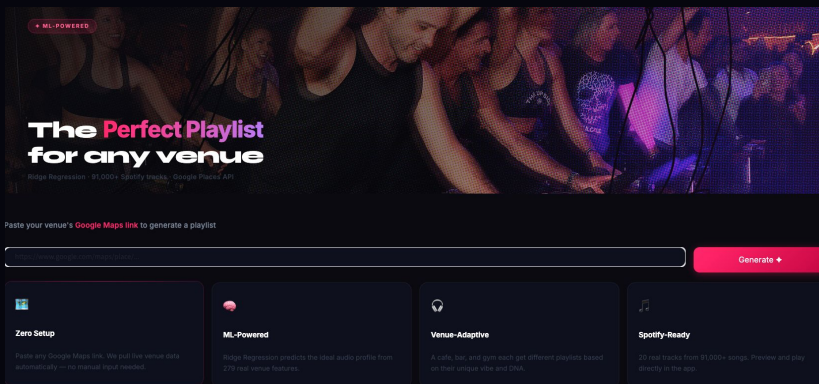
Ridge Regression



7-dim Audio Profile



20-track Playlist



TARGET USERS

- ▶ Restaurant managers
- ▶ Bar & club owners
- ▶ Hotel operators
- ▶ Retail store managers

KEY BENEFITS

- ▶ No music expertise needed
- ▶ Generate Playlist in < 1 second
- ▶ Real-time busyness adjustment
- ▶ Free alternative to B2B services

Technical Focus

- ▶ Ridge: 98.6% Top-5 Accuracy
- ▶ Mean Squared Error (MSE)
- ▶ Cosine Similarity
- ▶ Top-K Genre Retrieval Accuracy

Methods, Novelty & Social Impact

TWO ML MODELS — NumPy from scratch

Ridge Regression — closed-form solution $W^* = (X^T X + \lambda I)^{-1} X^T Y$

- interpretable weights, robust to correlated one-hot columns in 279-dim space.

Neural Network (128→64→7, ReLU/Sigmoid)

- captures non-linear feature interactions (e.g. live_music × price_level)

WHAT'S NOVEL

Google Maps venue metadata
as a direct acoustic predictor

Two-stage architecture: predict
audio profile → retrieve songs

SOCIAL IMPACT

- ▶ **Democratizes curation** — free data-driven music tool for small venues

ETHICAL CONCERNS

- ▶ **Genre bias risk** — genre mappings may not represent diverse acoustic traditions equitably
- ▶ **Geographic scope** — trained on Manhattan only; may not generalize globally
- ▶ **Pseudo-label Circularity** — constructed by mapping primary_type to genre groups
- ▶ **Commercial Public Performance Use** — Needs authorization license to play the list in stores

Two Datasets: Venues + Music

DATASET 1 — MANHATTAN VENUES

4,484

Venues

41

Columns

175

Primary Types

Source: Google Maps Places API v1 (April 2026)

NUMERICAL

Rating (1–5) · Coordinates

CATEGORICAL

Primary type · Price (4 tiers) · Neighbourhood (33)

BOOLEAN (18)

live_music · outdoor_seating · allows_dogs...

FREE TEXT

Google generative summary · User review summary

DATASET 2 — SPOTIFY TRACKS

91,271

Filtered Tracks

112

Genre Centroids

7

Audio Features

Source: Kaggle Spotify Tracks Dataset (114,000 → filtered popularity ≥ 10)

TARGET SPACE — 7 AUDIO DIMENSIONS

liveness

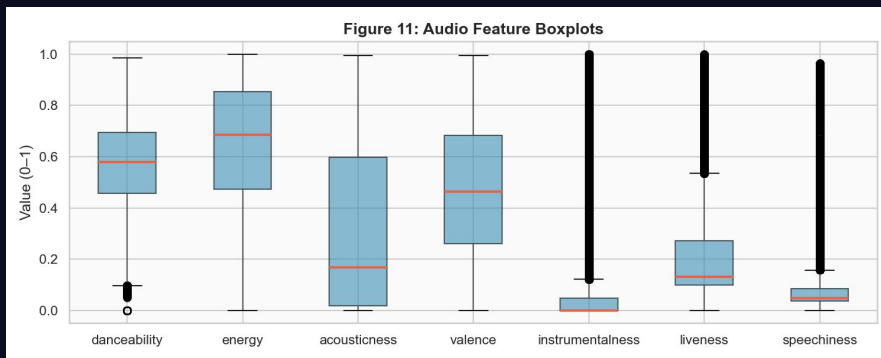
danceability

speechiness

acousticness

valence

instrumental



Preprocessing & Pseudo-Label Construction

PREPROCESSING → 4,484 × 279 FEATURE MATRIX

- ① Boolean encoding: 1=true, 0=false, -1=unknown — preserves missingness as signal (50-73% of high-signal flags unreported)
- ② Rating: 2% missing → median imputation (4.5). Price: ordinal 1-4 + separate binary missing indicator column
- ③ Categorical one-hot: 33 neighbourhoods + 175 primary types = 208 binary features. Unseen values at inference → zero vector
- ④ Text: review + generative summary → TF-IDF (unigrams+bigrams, sublinear TF, L2 norm) → Truncated SVD → 50 dense dims
- ⑤ Min-max normalize all features to [0, 1]

RESULT

1+1+1+18+33+175+50 = 279 features total

PSEUDO-LABEL CONSTRUCTION (no ground truth)

- ①
 1. Map primary type → genre group
 2. italian_restaurant → jazz / classical / soul
 3. bar → jazz / blues / rock
 4. Take mean Spotify audio profile as base target

Modulate with flags + price:

- ②
 1. live_music=True: +15% toward jazz/blues/folk
 2. outdoor_seating=True: +8% toward folk/acoustic
 3. Price level 4: -0.04 energy, +0.06 acousticness

RESULT

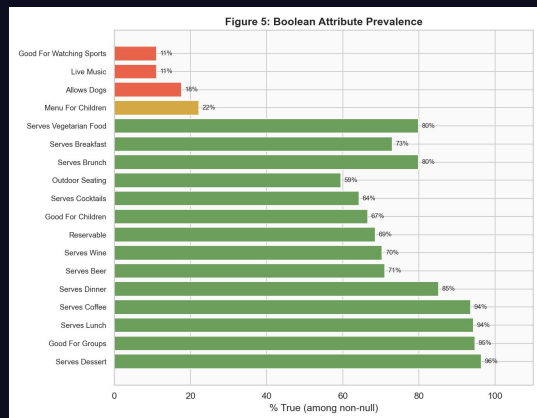
4,484 × 7 label matrix, clipped to [0, 1]

80/20 train/test, stratified on primary_type, seed=42

Labels built before split — no leakage

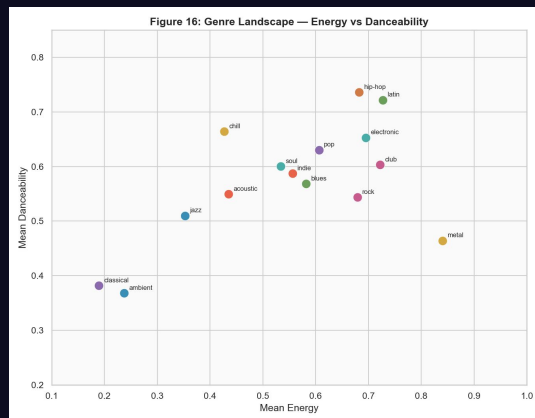
3 Findings That Informed Design

FINDING 1 — BOOLEAN PREVALENCE (Fig 5)



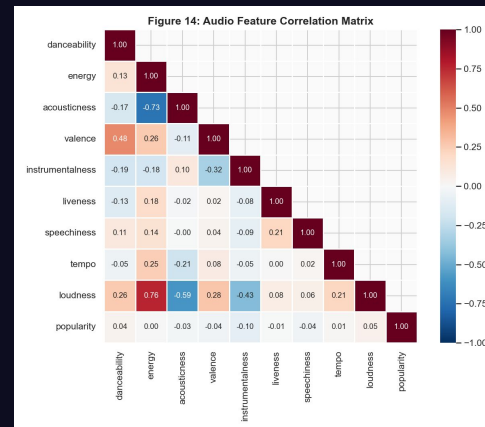
★ Rare = strongest differentiators
 → 50-73% missing → ternary encoding (1/0/-1)

FINDING 2 — GENRE SCATTER (Fig 16)



Distinct spatial separation
 → validates centroid retrieval architecture

FINDING 3 — CORRELATION MATRIX (Fig 14)



Audio features is correlate with each other
 → some more so than others

Problem Formulation & Equations

Task: $x \in \mathbb{R}^{279} \rightarrow \hat{y} \in [0,1]^7$. Regression (not classification): small differences in predicted audio features translate to meaningfully different genre matches.

METHOD 1: MULTI-OUTPUT RIDGE REGRESSION

$$\mathcal{L}(W) = \|XW - Y\|_F^2 + \lambda \|W\|_F^2$$

where $X \in \mathbb{R}^{n \times (d+1)}$ is the feature matrix augmented with a bias column of ones, $Y \in \mathbb{R}^{n \times 7}$ is the pseudo-label matrix, and $W \in \mathbb{R}^{(d+1) \times 7}$ contains the weights for all 7 output dimensions. The closed-form solution is:

$$W^* = (X^\top X + \lambda I)^{-1} X^\top Y$$

Bias row excluded from regularization. L2 shrinks correlated one-hot cols jointly without eliminating them. $W^*_{:,j}$ directly interpretable per audio dimension.

METHOD 2: TWO-LAYER NEURAL NETWORK

$$z^{(1)} = XW^{(1)} + b^{(1)}, \quad a^{(1)} = \max(0, z^{(1)})$$

$$z^{(2)} = a^{(1)}W^{(2)} + b^{(2)}, \quad a^{(2)} = \max(0, z^{(2)})$$

$$\hat{y} = \sigma(a^{(2)}W^{(3)} + b^{(3)}), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with layer dimensions $W^{(1)} \in \mathbb{R}^{279 \times 128}$, $W^{(2)} \in \mathbb{R}^{128 \times 64}$, $W^{(3)} \in \mathbb{R}^{64 \times 7}$. The training objective is MSE over all 7 outputs:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \|\hat{y}_i - y_i\|^2$$

Sigmoid bounds output to [0,1] without clipping. Captures live_music × price_level → jazz vs rock conditional that Ridge (linear model) cannot represent.

METHOD 3: K-MEANS (INTERPRETIVE BASELINE — K=25 by silhouette score)

$$\mathcal{L} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Unsupervised — validates natural venue groupings without pseudo-label dependency. Cluster-level granularity too coarse for deployment → interpretive tool only.

Inputs, Outputs & Playlist Retrieval

MODEL INPUTS → OUTPUTS

INPUT (all 3 models)

279-dim feature vector

rating · price · 18 bool · 33 nbhd · 175 type · 50 text SVD



OUTPUT (Ridge & NN)

7-dim audio profile $\hat{y} \in [0,1]^7$

one predicted value per Spotify audio feature

K-Means → cluster centroid profile used as prediction

PLAYLIST RETRIEVAL — 3 STEPS

1

Nearest Genre Lookup

Compute Euclidean distance from $\hat{y}$ to all 112 genre centroids → select top-5 nearest genres.

2

Weighted Sampling

Sample 20 tracks proportional to inverse distance weights. Within each genre, sampling probability $\propto$ Spotify popularity (favors recognizable songs).

3

Energy Arc Ordering

Sort 20 tracks: low energy → midpoint peak → cool-down. Mimics DJ curation for a continuous venue service period.

Evaluation Metrics & Experimental Results

① MSE

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i^{\text{true}} - \mathbf{y}_i^{\text{pred}})^2$$

Comparable across 7 dims since targets $\in [0,1]$. Lower = better. Baseline: 0.004124 → Ridge: 0.000281

② Cosine Similarity

$$\text{CosSim} = \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{y}_i^{\text{true}} \cdot \mathbf{y}_i^{\text{pred}}}{\|\mathbf{y}_i^{\text{true}}\| \cdot \|\mathbf{y}_i^{\text{pred}}\|}$$

Evaluates profile shape not magnitude. Critical for low-variance dims (liveness $\sigma=0.045$)

③ Top-K Genre Accuracy

$$\frac{1}{n} \sum_{i=1}^n 1 \left[\arg \min_j \|\mathbf{c}_j - \mathbf{y}_i^{\text{true}}\| \in \text{TopK}(\mathbf{y}_i^{\text{pred}}) \right]$$

Most directly reflects end-user value. Does the right genre appear in the retrieval set?

TEST SET RESULTS — BEST vs. WORST PERFORMING ALGORITHMS

Model	Test MSE ↓	Cosine Sim ↑	Top-3 Acc ↑	Top-5 Acc ↑	Status
Mean Baseline — WORST	0.004124	0.9874	29.2%	36.3%	▼ Worst
K-Means (K=25)	0.001184	0.9964	85.3%	88.3%	—
Neural Network	0.000255	0.9992	97.8%	98.3%	Strong
Ridge ($\lambda=0.1$) ★ DEPLOYED	0.000281	0.9991	98.0%	98.6%	★ Deployed

★ Ridge: Top-5 gap = 0.3pp (<3% margin) → tiebreaker: single matrix-mult. inference + directly interpretable weight matrix per audio dimension

Training Procedure, Hyperparameter Tuning & Regularization

TRAIN / VAL / TEST SPLIT 64/20/16 (stratified, seed=42)

Train 64% (3,587), Validate 16%

Test 20% (897)

Train 2,869

Val 718

- ▶ All transformers fit on train subset only
- ▶ Pseudo-labels built before split — no leakage
- ▶ Test set never touched until final evaluation

OVERFITTING & UNDERFITTING PREVENTION

- ▶ Ridge: L2 penalty inherently prevents overfit; train/val MSE gap < 0.00005 at $\lambda=0.1$ (confirms no underfit)
- ▶ NN: Dropout (rate 0.2) + Early stopping (patience=20, restores best-epoch weights)
- ▶ No train/val loss divergence in selected configuration

HYPERPARAMETER TUNING — GRID SEARCH

Ridge λ Sweep — Validation MSE (optimal $\lambda=0.1$ highlighted)



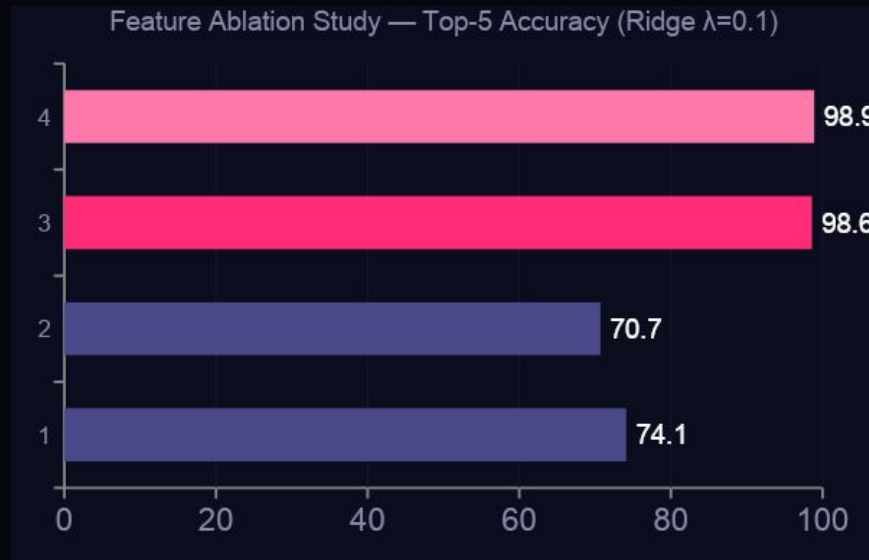
NEURAL NETWORK — 18 CONFIGS

$\alpha \in \{0.001, 0.005, 0.01\} \cdot \text{batch} \in \{32, 64, 128\} \cdot \text{dropout} \in \{0.0, 0.2\}$

Max 500 epochs. Best config: $\alpha=0.01$, batch=32, dropout=0.0

K-Means: $K \in \{10, 15, 20, 25, 30\} \rightarrow K=25$ by silhouette score

Results Charts & Feature Ablation Study



KEY RESULT

- ▶ All learned models vastly outperform baseline (36.3% → 98.6%)
- ▶ Ridge = NN in accuracy; faster + interpretable → deployed

ABLATION INSIGHT

- ▶ `primary_type` alone achieves 98.6% — the dominant predictor
- ▶ Boolean and text add minimal gain — consequence of pseudo-label circularity

App Overview, Model Selection & Target Users

APPLICATION OVERVIEW

6-page Streamlit app: Home · Statistics · Playlist · About · Favorites · Recent

- ▶ Pipeline cached via `@st.cache_resource` — no redundant model calls
- ▶ Session state manages all page transitions

MODEL SELECTION — 3-TIER DECISION

- ① Top-5 Accuracy: Ridge 98.6% vs NN 98.3% — gap 0.3pp < 3% → INCONCLUSIVE
- ② Cosine Similarity: 0.9991 vs 0.9992 — also within margin
- ③ TIEBREAKER → RIDGE WINS: single matrix mult. = sub-second inference + weight matrix directly interpretable per audio dimension

TARGET POPULATION & BENEFITS

CURRENT ALTERNATIVES

Manual curation (requires expertise) or B2B services (expensive, standardized content)



ATMOSOUND

Free · Instant · Data-driven
· Venue-specific

- ▶ Target: venue owners, bar managers, hotel operators — need acoustically appropriate music without music expertise
- ▶ Input: Google Maps listing they already have
- ▶ Output: curated 20-track playlist in < 1 second with audio profile visualization

BUSYNESS SLIDERS — REAL-TIME OVERRIDE

6 sliders (7am–2am). Compute $\bar{d} = (\bar{b} - 0.5) \times 0.35$
 $\text{energy} += \bar{d} \cdot \text{danceability} += 0.8\bar{d} \cdot \text{acousticness} -= 0.7\bar{d} \cdot \text{valence}$
 $+= 0.5\bar{d}$

Regenerates playlist without re-running Ridge — sub-second responsive.

Lets operators adjust when current atmosphere $\neq$ historical data.

Summary, Implications & Future Work

PROJECT GOALS & KEY RESULTS

- ▶ Google Maps venue metadata is a viable acoustic predictor — first system to demonstrate this
- ▶ Ridge Regression (NumPy): 98.6% Top-5 vs 36.3% baseline — 14.7× improvement
- ▶ 3-tier selection: Ridge = NN in accuracy; faster + interpretable → deployed
- ▶ Feature ablation: primary_type dominant (categorical-only = 98.6%); flags & text marginal
- ▶ Streamlit app: busyness sliders + Spotify embeds → free tool for operators

IMPLICATIONS FOR USERS & SOCIETY

- ▶ Small venues get a free, data-driven alternative to expensive B2B curation services
- ▶ Busyness sliders allow real-time control when atmosphere ≠ historical baseline
- ▶ Publicly available venue metadata is a viable acoustic signal — opens new direction in context-aware recommendation
- ▶ Reduces cost of entry for music curation across hospitality and retail

A CAVEAT

- ▶ Venue data quality on Google Maps varies significantly. The model's predictions are only as good as the underlying data.
- ▶ Model trained on Manhattan only — neighbourhood encoding fails outside vocabulary

FUTURE PLANS — LESSONS APPLIED

- ▶ Human-annotated ground truth to fix pseudo-label circularity (survey venue managers)
- ▶ Live Spotify API for a dynamic, current song pool
- ▶ Extend training beyond Manhattan — reduce neighbourhood encoding bias
- ▶ Time-of-day as structured model input, not just busyness sliders

THANK YOU